Alkanes

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Alkanes

Electrophile

In a reaction between two organic molecules, it is difficult to specify the reagent and the substrate. Ordinarily, the one with full charge is more reactive and may be considered as the reagent. Thus, $C_2H_5O^{\ominus}$ is the reagent in the reaction given below:

$$\underset{\text{Substrate}}{R{-}X} + \underset{\text{Reagent}}{C_2H_5O^{\ominus}Na^{\oplus}} \rightarrow \underset{\text{Product}}{C_2H_5OR} + NaX$$

Reagents are of Two Types

The one which attacks the positive site in the molecule is called the nucleophile or nucleophilic reagent and the one which attacks the negative site in the molecule is called the electrophile or electrophilic reagent.

1. Electrophile:

Electron deficient species which are attracted towards electron. Electrophiles are positively charged or neutral species which have vacant orbital.

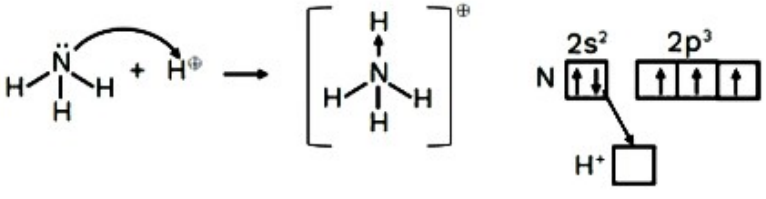
Types of Electrophile:

(A) Cations: R^{+}, $(CH_3^{\oplus}, CH_3{-}CH_2^{\oplus})$ $H^{\oplus}$, $\overset{\oplus}{N}O_2$, $\overset{\oplus}{N}O$, $\overset{\oplus}{S}O_3H$, $H-\overset{\oplus}{C}=O$, $R-\overset{O}{\overset{\|}{C}}{}^{\oplus}$,

$\overset{\oplus}{C}H=NH$, $Cl^{\oplus}$, $Br^{\oplus}$

$NH_4^{\oplus}$	and	$H_3O^{\oplus}$
IA	and	IIA cations
Na^{+}, K^{+}		Mg^{2+}, Ca^{2+}

* Do not work as electrophile because they have complete octet.



(B) Neutral Electron:

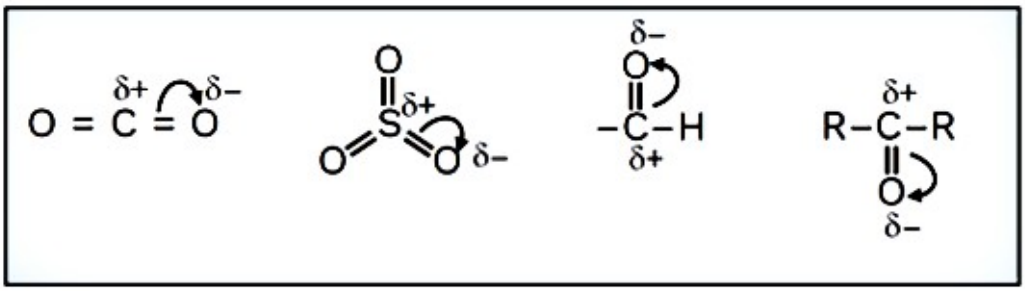
$AlCl_3$, BF_3, BCl_3, BH_3, $SiCl_4$, PCl_3, PCl_5 (Vacant orbital), $SnCl_2$, $SnCl_4$, $SbCl_5$

All Lewis Acids are $E^{\oplus}$

$E^{\oplus}$ deficient species

(C) Polar π-bond:

Example:



(D) Induced Dipole Interaction :

Example : Br_2, Cl_2, I_2

ILLUSTRATIONS

Q1 **An electrophilic reagent is:**

(A) Electron rich species **(B) Electron deficient species**

(C) A Lewis acid **(D) Negatively charged species**

Sol: B

Q2 **A nucleophile is:**

(A) Electron-rich species **(B) Electron deficient species**

(C) a Lewis acid **(D) Positively charged species**

Sol: A

Q3 **Which of the following will act as an electrophilic reagent?**
(A) H_2O **(B) OH^-** **(C) NO_2^+** **(D) None**

Sol: C

Q4 **Which one is not an electrophile in given below?**
(A) $AlBr_3$ **(B) BF_3** **(C) $SnCl_4$** **(D) NH_3**

Sol: D

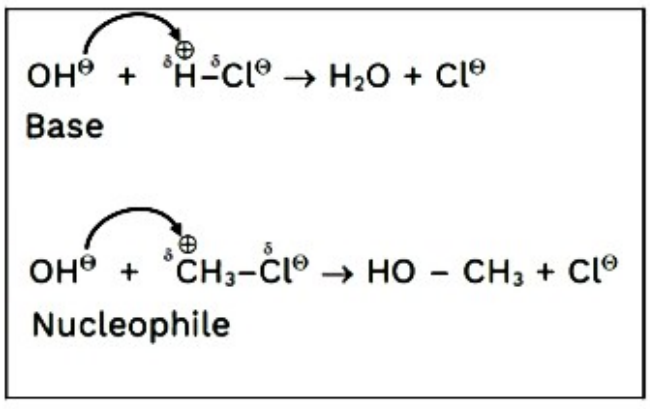
Q5 **Which among the following is not a Lewis acid?**
(A) BF_3 **(B) $AlCl_3$** **(C) NH_3** **(D) Hg^{2+}**

Sol: C

Q6 **Which of the following is an electrophile?**
(A) NH_3 **(B) ROH** **(C) ^-OR** **(D) BCl_3**

Sol: D

2. **Nucleophiles:** Electron rich species which attack on electron deficient centre [except H^+] attracted towards electron deficient centre.



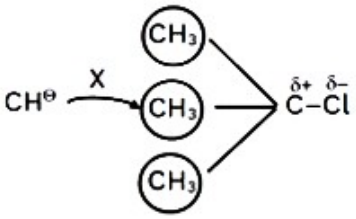
Points to remember

All anions and lone pair containing substance are all Nucleophiles.

Nucleophile	Base
● Electron rich species which attack on electron deficient centre [except $H^{\oplus}$]	● Electron rich species which accept $H^{\oplus}$ ions
➡ $OH^{\ominus} + \overset{\delta+}{C}H_3 - \overset{\delta-}{Cl} \rightarrow HO-CH_3 + Cl^{\ominus}$	➡ $OH^{\ominus} + H^{\oplus} - Cl^{\ominus} \rightarrow H_2O + Cl^{\ominus}$

Nucleophilicity	Basicity
Kinetically controlled Attacking speed high → Strong $Nu^{\ominus}$ Attacking speed low → Weak $Nu^{\ominus}$	Thermodynamically controlled that is How much fast accept $H^{\oplus}$ ions form stable product.
As steric hinderance ↑, nucleophilicity decreases	Steric hinderance have no effect on basicity as acid-base reaction take places.

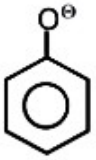
$(Nu^{\ominus} \downarrow)$



Types of Nucleophile:

(A) All anionic species :

(i) $R^{\ominus}$ ($CH_3^{\ominus}$, $CH_3-CH_2^{\ominus}$) (ii) $OH^{\ominus}$, $NO_2^{\ominus}$ (iii) $F^{\ominus}$, $Cl^{\ominus}$, $Br^{\ominus}$, $I^{\ominus}$

(iv) $CN^{\ominus}$, $SH^{\ominus}$, $R-O^{\ominus}$ (v)  , $CH_3-\overset{O}{\overset{\|}{C}}-O^{\ominus}$

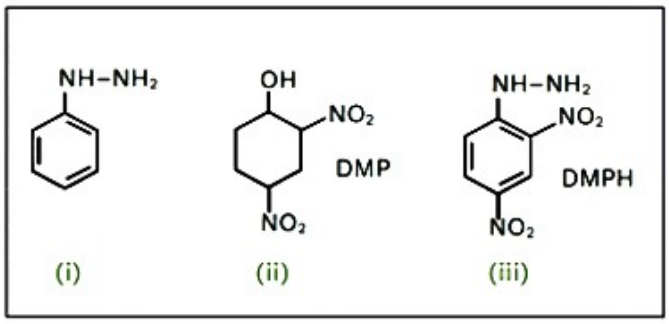
(B) All lone pair containing:

(i) $H_2\ddot{O}$, $R-\ddot{O}H$

(ii) $\ddot{N}H_3$, $R-\ddot{N}H_2$, $R-\ddot{N}H-R$, $R-\underset{\underset{R}{|}}{\ddot{N}}-R$

(iii) $R-\ddot{S}H$, $\ddot{P}H_3$, $\ddot{P}Cl_3$

(iv) $H-O-O-H$, NH_2-NH_2



Points to remember

Some species act as both electrophile nucleophile called as ambiphiles :

Electrophiles as well Nucleophile Vacantorbital Lone pair.

(C) π-bond: Alkene, alkyne and phenyl

Example:

$R-CH=CH_2$, $R-C\equiv CH$

Factors affecting Nucleophilicity:

(A) Electron Density :

➡ Electron density ↑ Nucleophilicity ↑

(i) $OH^{\ominus} > H_2\ddot{O}$ (ii) $NH_2^{\ominus} > \ddot{N}H_3$

(iii) $SH^{\ominus} > H_2\ddot{S}$

(B) Nucleophilicity in Periodic Table :

(i) In a period :

Electronegativity ↓ Nucleophilicity ↓

Nucleophilicity : $CH_2^{\ominus} > NH_2^{\ominus} > OH^{\ominus} > F^{\ominus}$

Basic Strength : $CH_2^{\ominus} > NH_2^{\ominus} > OH^{\ominus} > F^{\ominus}$

(ii) In a group :

In a group, size of species/cation/anion changes so solvent effect is applicable.

We will discuss solvent effect in groups at nucleophilicity in halogen derivatives before SN2 reaction .

ILLUSTRATIONS

Q7 **Which one is not a nucleophile in below?**
(A) NH_3 **(B) BF_3** **(C) $-OH$** **(D) HSO_3^-**

Sol: B

Q8 **A nucleophile must necessarily contain**
(A) an overall positive charge **(B) an overall negative charge**
(C) a lone pair of electrons **(D) an unpaired electron**

Sol: C

Q9 **Which of the following is an electrophile?**
(A) RNH_2 **(B) H_2O** **(C) $:CN^-$** **(D) $:CH_2$**

Sol: D

3. Reaction Intermediates :

Reaction intermediates are basically related to species formed inbetween reaction. Bond breaking is either hetrolytic or homolytic.

(A) Homolysis (Bond breaking in equal sharing basis)

Low electronegative difference leads to formation of radical inter mediate result due to homolytic bond breaking.

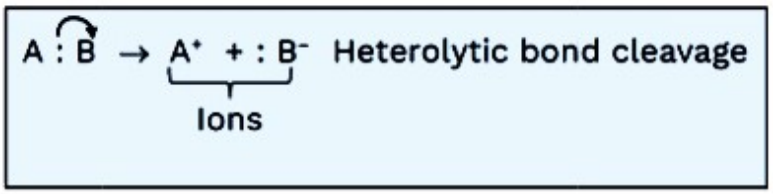


Definitions

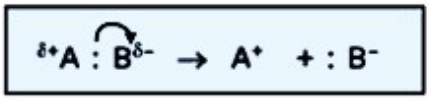
Reactive, short lived, high energy, unstable species, those are formed in the course of organic reactions are called reaction intermediates.

(B) Hetrolysis (Unequal distribution of electrons):

Bond breaking in such a manner that bond electron shift permanently to one fragment charged fragment produce (positive and negative) ions.



Bond polarity is key reason behind hetrolysis.

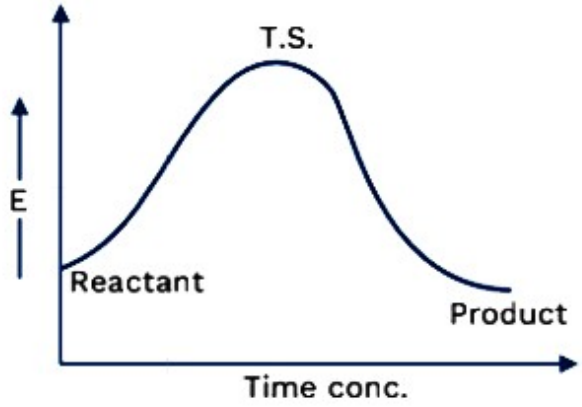


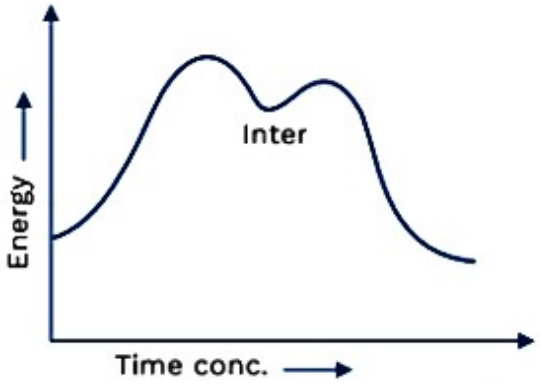
Energy Profile Diagram

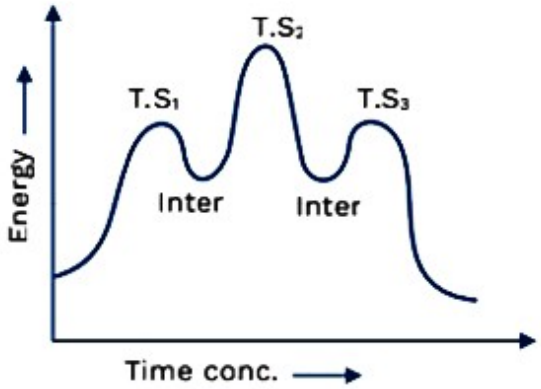
- Energy profile diagram is started from reactant to product in which transition state form.

Let's understand

Those diagrams which explain the conversion of reactant into product with energy is known as energy profile diagram.







Number of transition state = number of step = 1 + intermediate

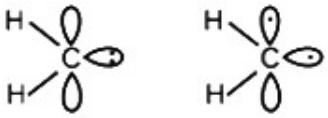
- The rate determining step is a slow step in which maximum activation energy is required.
- To count activation energy difference between reactant and highest energy transition state.
- Intermediate means those species which form between two Transition state.

Examples of intermediate:

1. Carbocation $CH_3^{\oplus}$

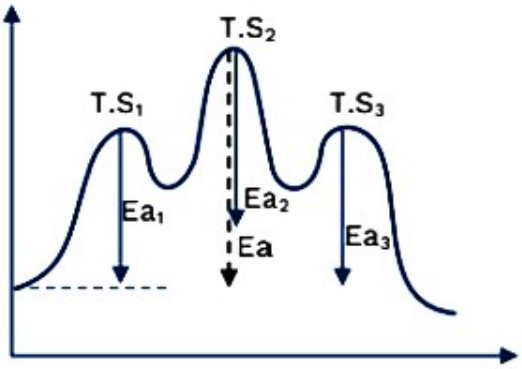
2. Carbanion $CH_3^{\ominus}$

3. Free radical $CH_3^{\bullet}$

4. Carbene 

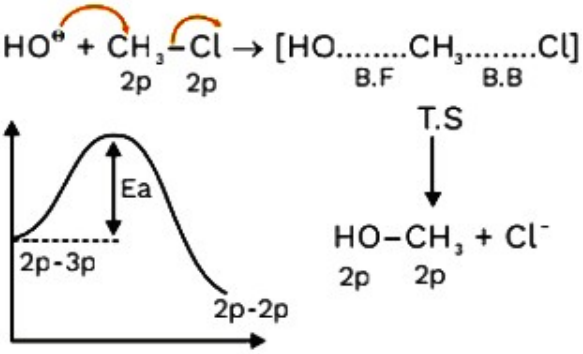
5. Nitrene 

6. Benzyne 

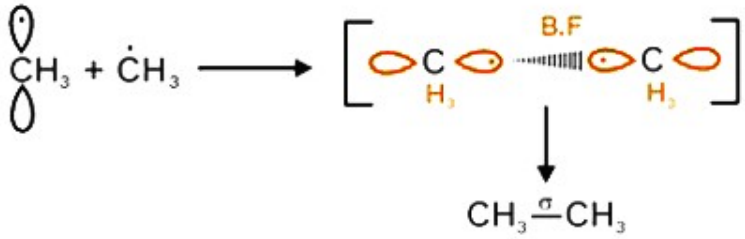


Transition State :

Those Species formed during a reaction which cannot be isolated or separated is known as transition state.



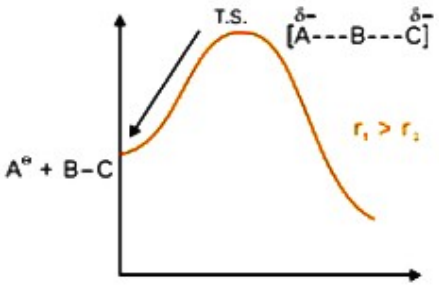
Exception :



Type of transition state :

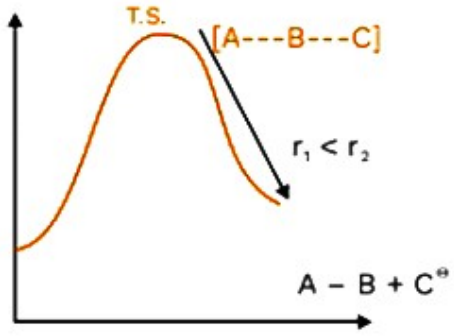
(A) Exothermic Transition State :

Transition state is close enough to energy to the reactants than to the products. Therefore, the transition state is having reactant character.



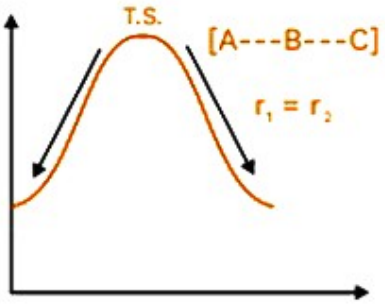
(B) Endothermic Transition State :

The transition state is closer enough to energy to the products than to the reactants. Therefore, Transition state is having product character.



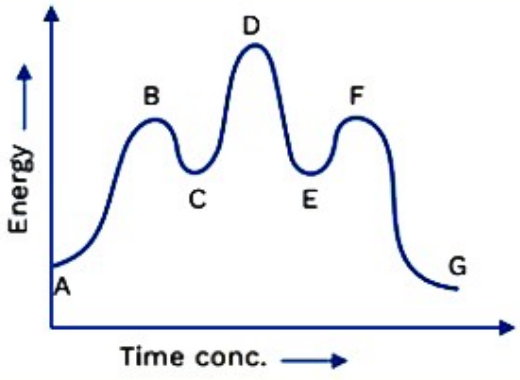
(C) Isoenthalpic Transition State :

Transition state is closer enough to both reactant and product which means both characters.



ILLUSTRATION

Q10



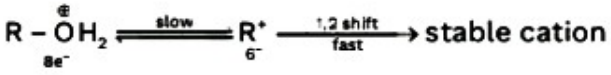
Sol: (i) Number of step =3
(ii) R.D.S of step=II
(iii) Number of transition state = 3(B.D.F)
(iv) Number of intermediate = 2 (C.E)
(v) B resembles with C
(vi) D resembles with C
(vii) F resemble with E
(viii) C resemble with B

Rearrangement of Carbocations

Let's understand

In a reaction if intermediate carbocation will form, then it may reaaranges. Only that carbocation will rearrange which can produce more stable species.

- Cation formation is slow step, but rearrangement is a fast step.



It can be done either by
(i) Shifting of hydride (H-), alkyl (R-), aryl (Ph-), bond, (1, 2 shifting)
(ii) Ring expansion (more strained ring to less strained ring)
(iii) Ring contraction

If the carbocation formed in a reaction is unstable, it rearranges to a more stable carbocation by 1, 2 shift.

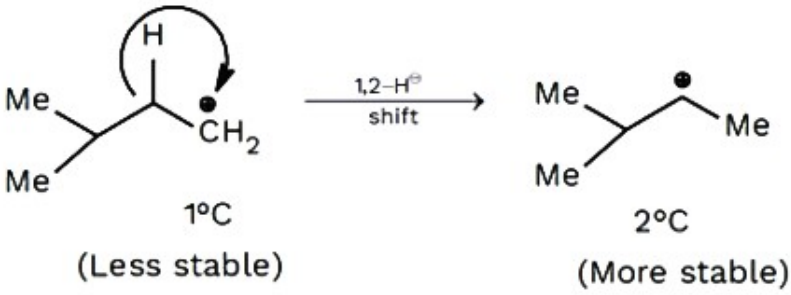
1, 2 $H^{\oplus}$ shift
1, 2 $CH_3^{\oplus}$ shift
1, 2 $Ph^{\oplus}$ shift

1, 2 alkyl shift
1, 2 ring expansion
1, 2 ring contraction

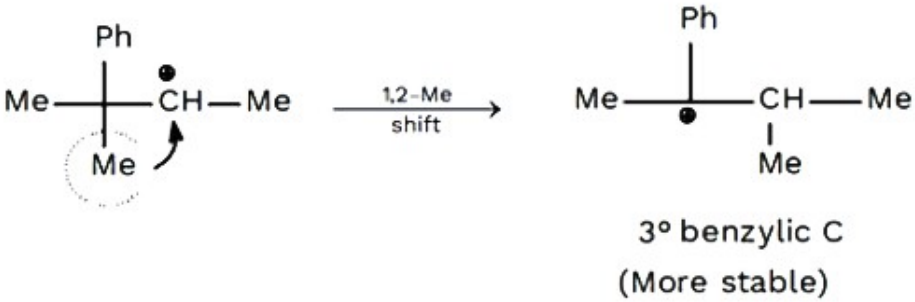
* All are fast step
* All are possible 1, 2 shift

Types of Shifting

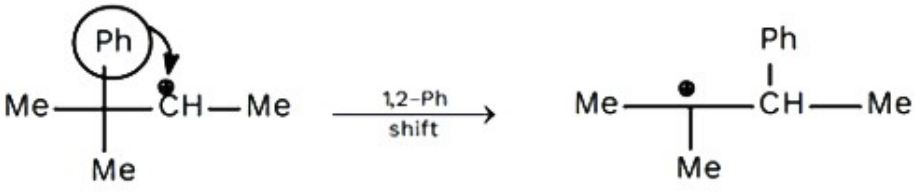
1. 1, 2 Hydride Shift



2. 1,2-Methyl shift and 1,2-phenyl shift



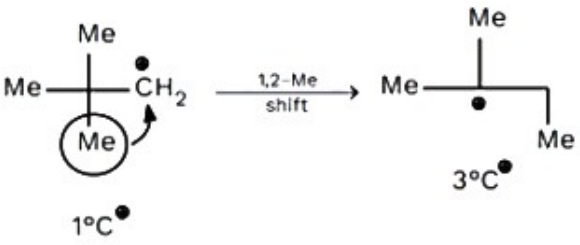
* 1st Possibility



* 2nd Possibility

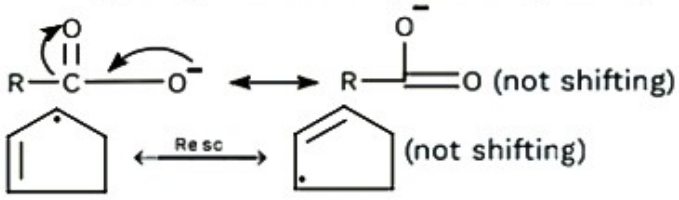
(Resonance is absent)

3. Neopentyl rearrangement



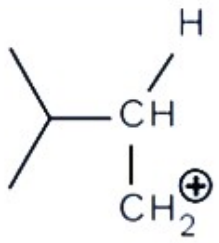
*For 1, 2 shifting cation must be more stable than before

*In alkyl, only cation shift (not anion, radical)

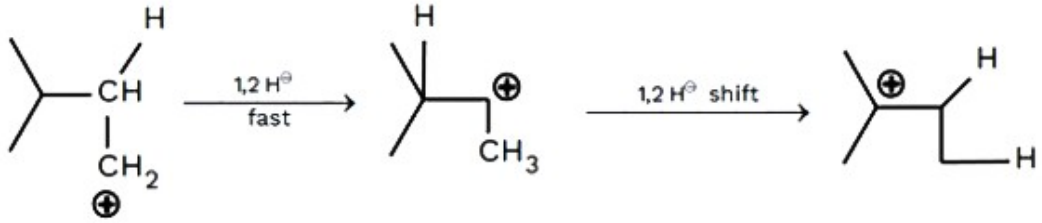


Count the number of shifting in the following:
Concept Building Exercis

Q11



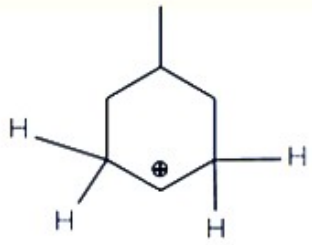
Sol:



2αH 4αH 8αH

* 2 times shift

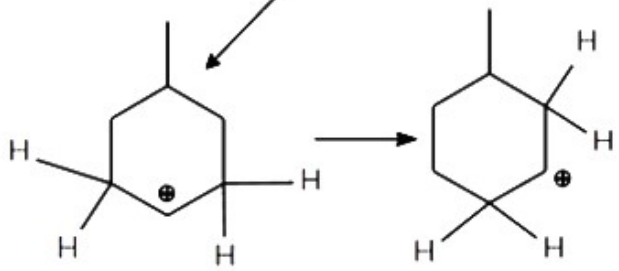
Q12



Sol:

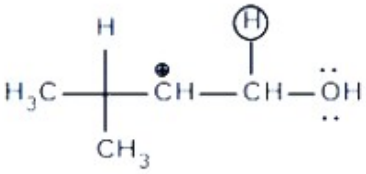
This is already in stable statel.

→ No change should take palce.

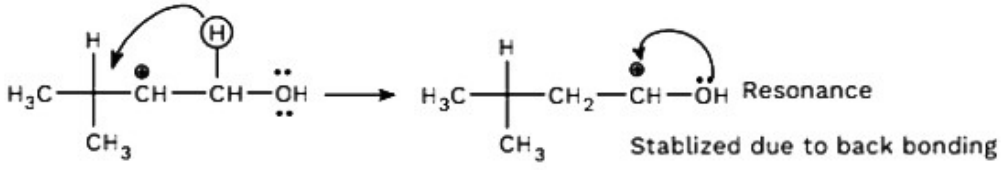


4αH 4αH

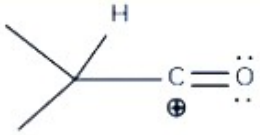
Q13



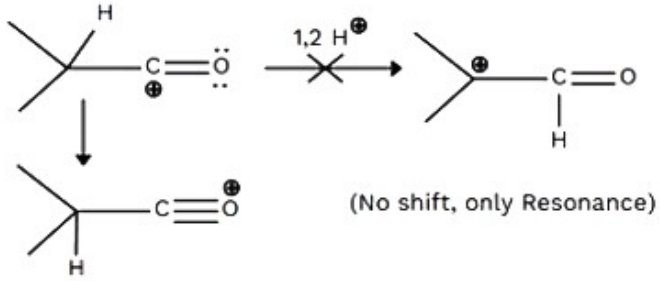
Sol:



Q14

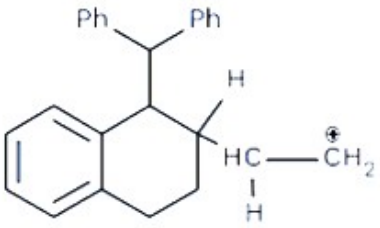


Sol:

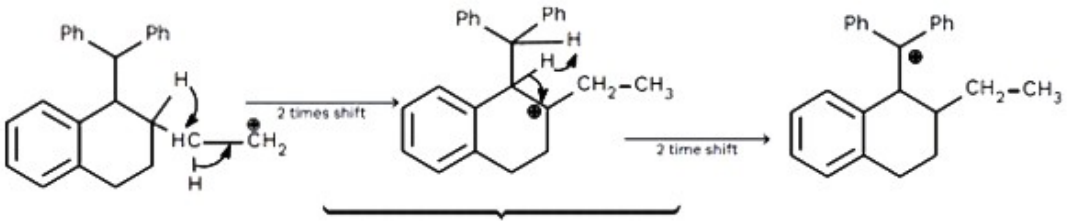


$8e^-$ (Octet compete cation)

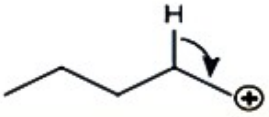
Q15



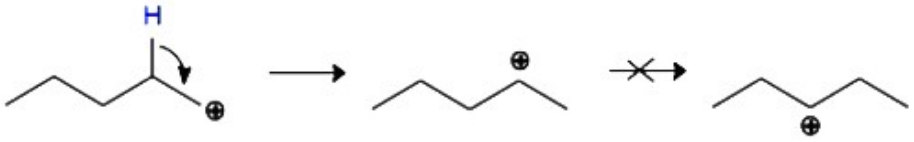
Sol:

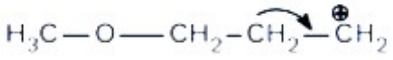


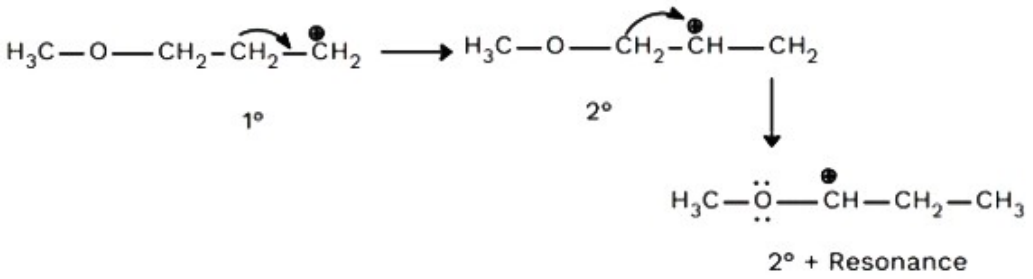
Q16

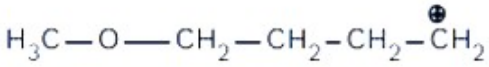


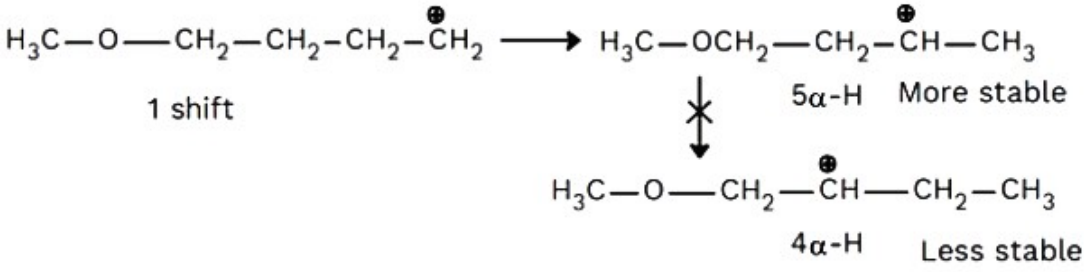
Sol:



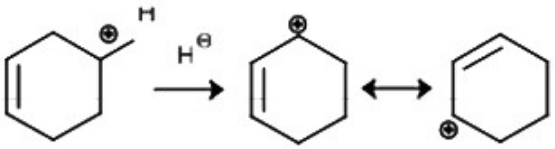
Q17 

Sol: 

Q18 

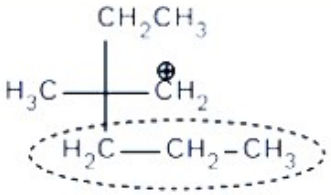
Sol: 

Q19 

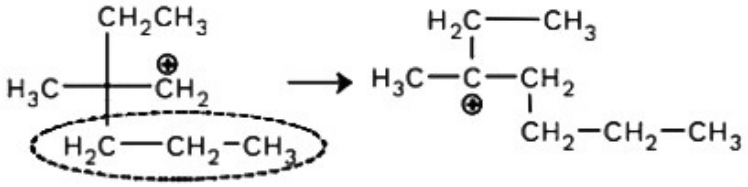
Sol. 

Resonance (not shifting)

Q20

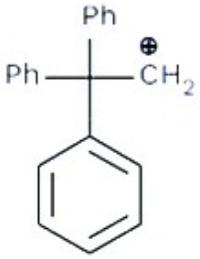


Sol.

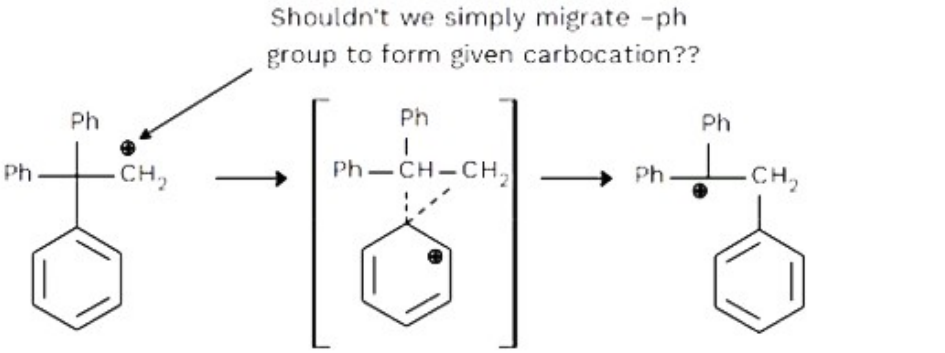


(more e^- density more $+I$), $6\alpha H$ (present)

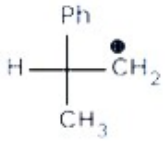
Q21



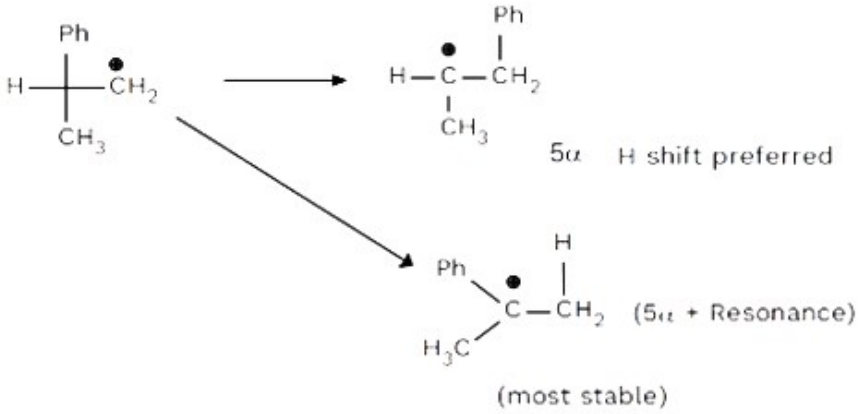
Sol.



Q22



Sol:



Ring Expansion
Carbocation out of Ring

3 membered → 4 membered → 5 membered → 6 membered ⟶⟷ 7 membered

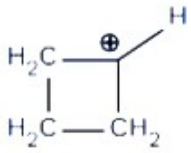
Possible

Points to remember

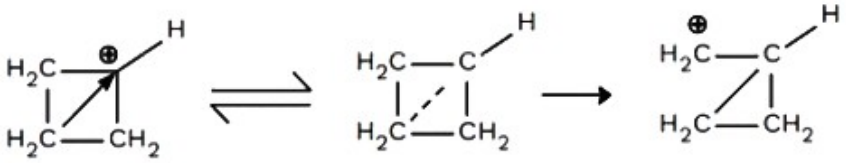
- For ring expansion, numbering start from carbocation to carbon ring and end to that carbon which is having more e^- donor.
- If cation in the ring then ring contraction occur.

Concept Building Exercis

Q23



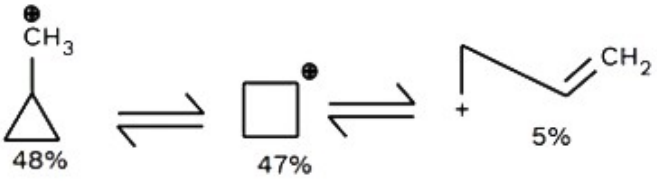
Sol:



Q24



Sol:

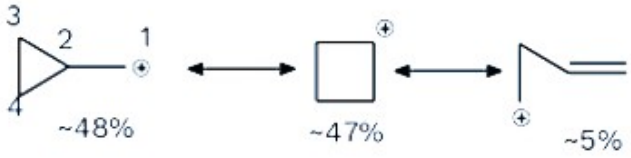


Demjanov equilibrium (Dancing Reso)

Know the Facts

CPM (Cyclopropyl methyl carbocation) :

3 member cyclopropyl carbocation highly strained due to ring angle strain. When positive charge present just outside of ring σ bond electron delocalization possible due to high relief in strain and lowering energy stabilize system. 2 σ electron and 3 'C' involved in delocalization.

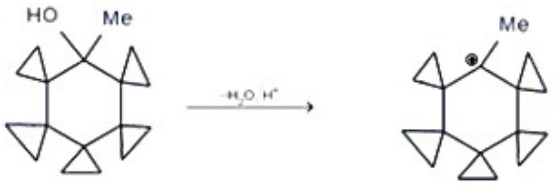


Bent orbital tend to shift towards vacant orbital of cationic carbon.

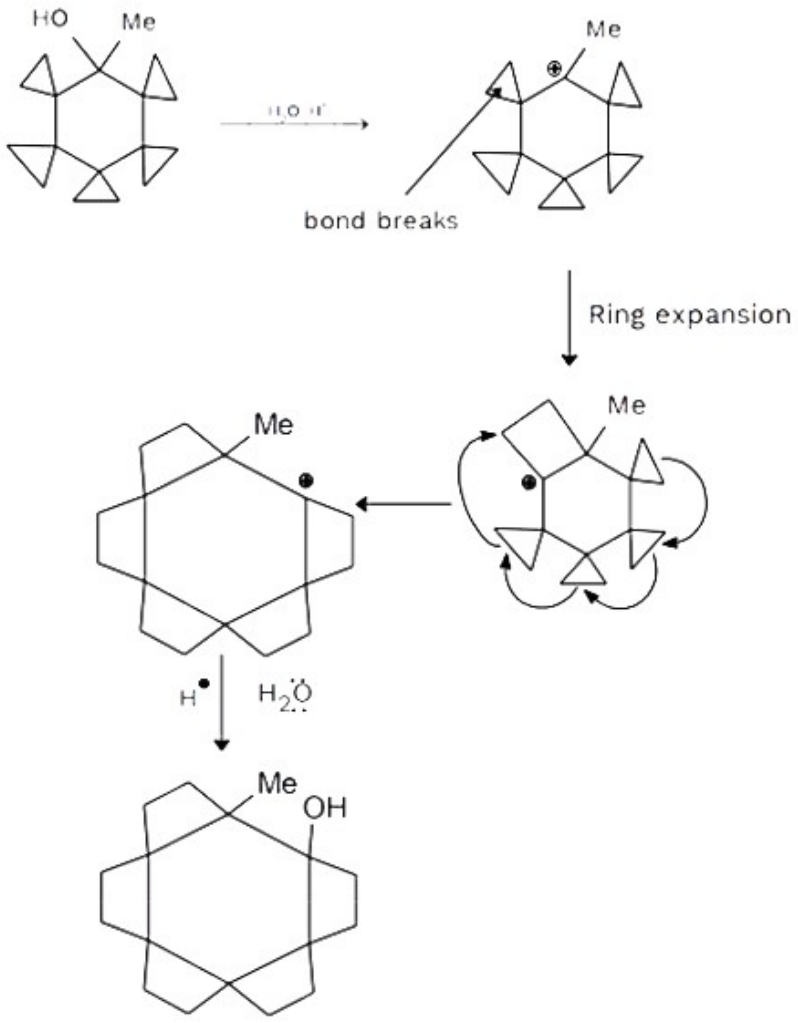


ILLUSTRATION

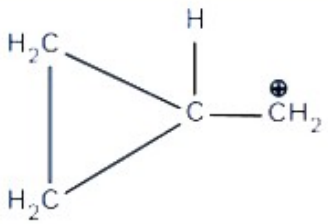
Q25



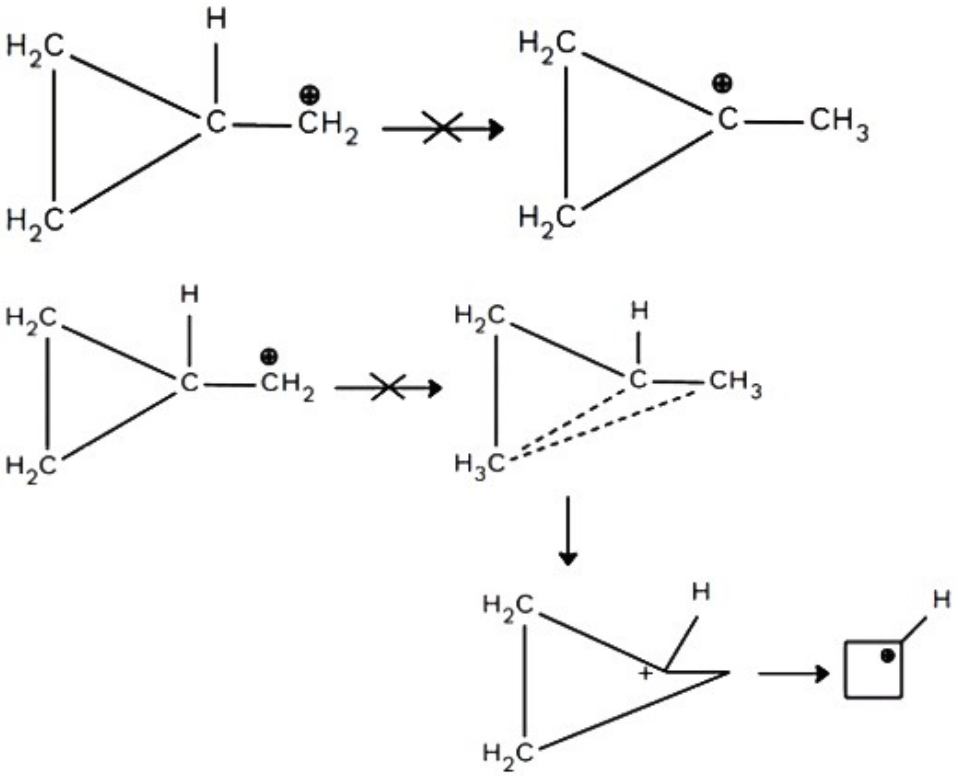
Sol.



Q26

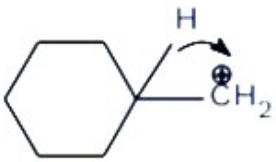


Sol.

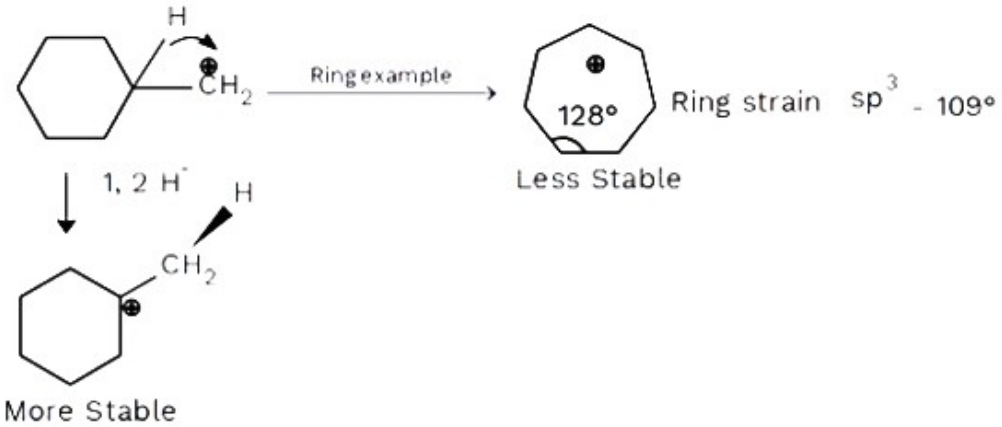


Though, we formed 4-membered as a result of ring expansion that carbocation outside 3 members ring is more stable.

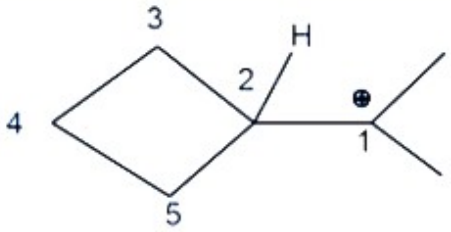
Q27



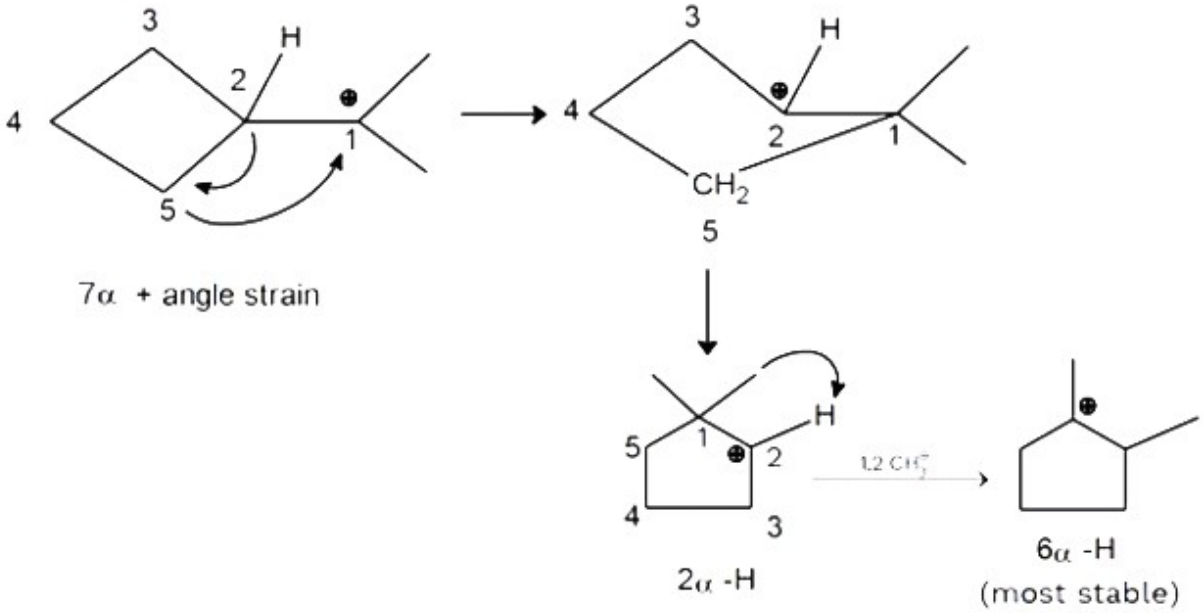
Sol.



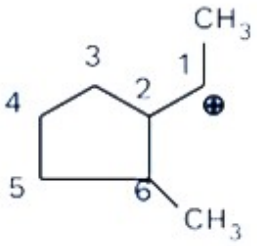
Q28


3
H
2
4
1
5

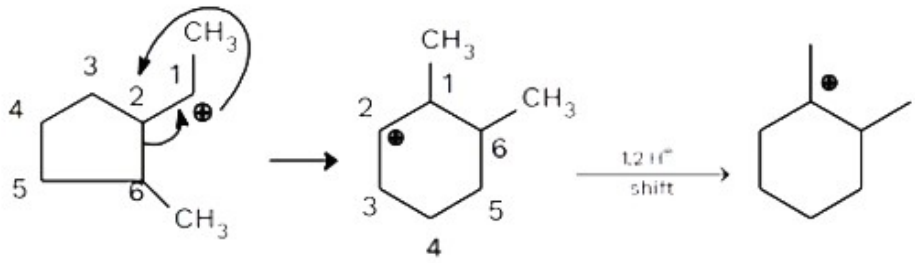
Sol.


3
H
2
4
1
5
7α + angle strain
3
H
4
2
1
CH2
5
H
5
1
2
4
3
1,2 CH3
2α -H
6α -H
(most stable)

Q29

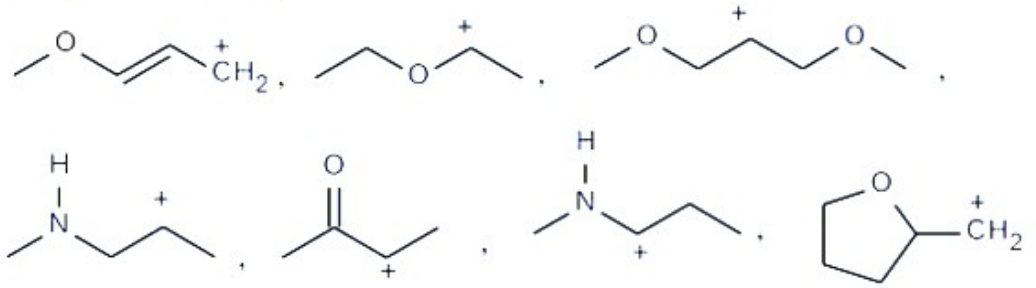

CH3
3
1
2
4
5
6
CH3

Sol.

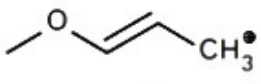


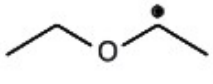
(Ring Expansion)

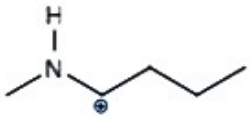
Q30 **How many of the following carbocations have stability greater than iso-propyl cation $(CH_3)_2\overset{+}{C}H$?**



Sol. 1,2,6

1.  Resonance stabilized.

2.  Resonance (Back bonding) stabilized

6.  Resonance (B.B.) stabilized

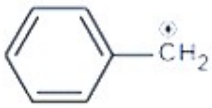
Passage

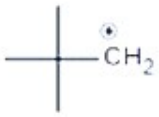
Under common reaction conditions, for greater stability a carbocation rearranges to another carbocation. Like, secondary carbocation will rearrange to a tertiary carbocation. It will not rearrange to a less stable primary carbocation. This generalization is not absolute, and because there is not a high energy barrier to the rearrangement of carbocations, rearrangement to a less stable cation can occur if it offers the chance to form a more stable product.

Q31 **Rearrangment of carbocation is possible for -**

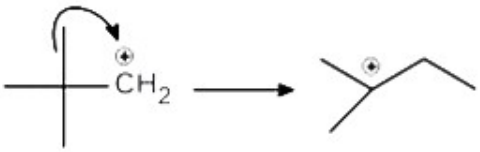
(A) 

(B) 

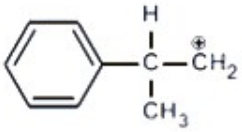
(C) 

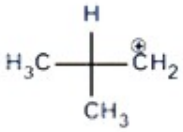
(D) 

Sol. D

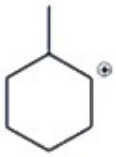


Q32 **In which of the following rate of cations rearrangement is maximum?**

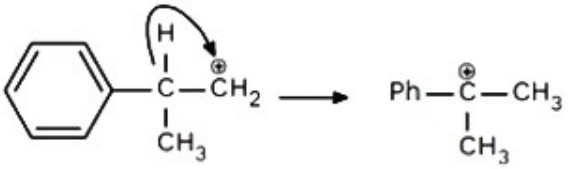
(A) 

(B) 

(C) 

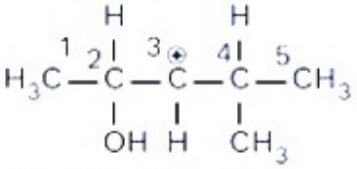
(D) 

Sol. A



As Rearranged carbocation is resonance stabilized by benzene ring.

Q33 **In the following cation, $\frac{H}{CH_3}$ that is most likely to migrate to the positively charged carbon is:**

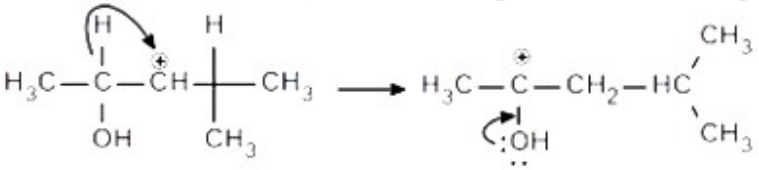


(A) CH_3 at C – 4 **(B) H at C – 4**

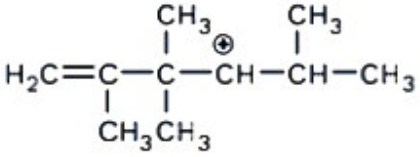
(C) CH_3 at C – 2 **(D) H at C – 2**

Sol. D

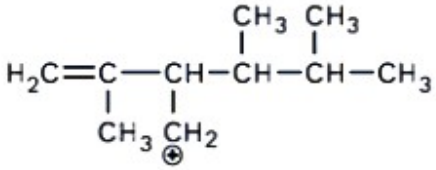
As when we move H for C-2 we get back bonding stabilized carbocation.



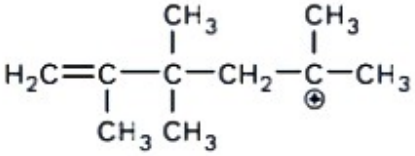
Q34 **The following carbocation rearranges to**



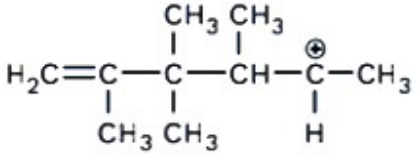
(A)



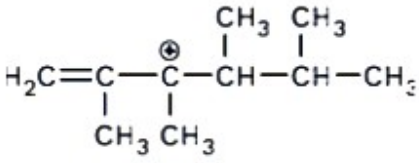
(B)



(C)



(D)



Sol. D

1,2-methyl shift results into more stable carbocation which is stable due to delocalisation and hyperconjugation.

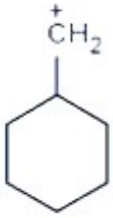
Q35 **Which of the following carbocation can undergo rearrangement?**

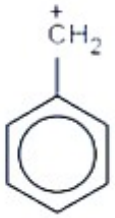
(a) 

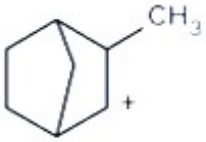
(b) 

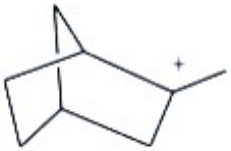
(c) 

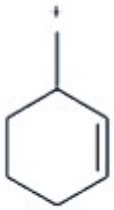
(d) 

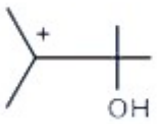
(e) 

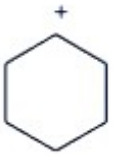
(f) 

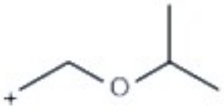
(g) 

(h) 

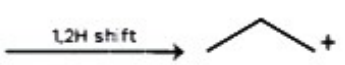
(i) 

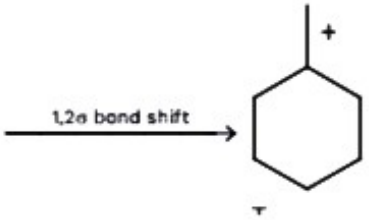
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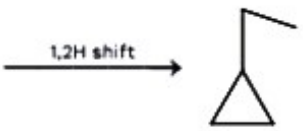
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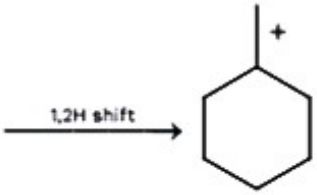
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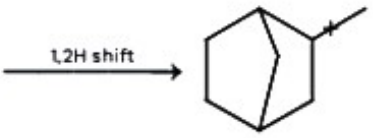
Sol. 8

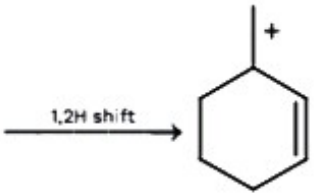
(b) 

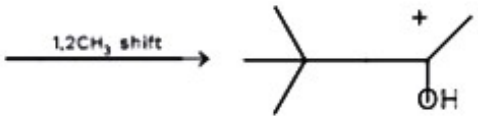
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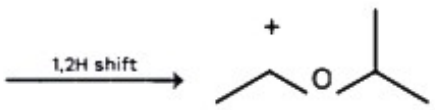
(d) 

(e) 

(g) 

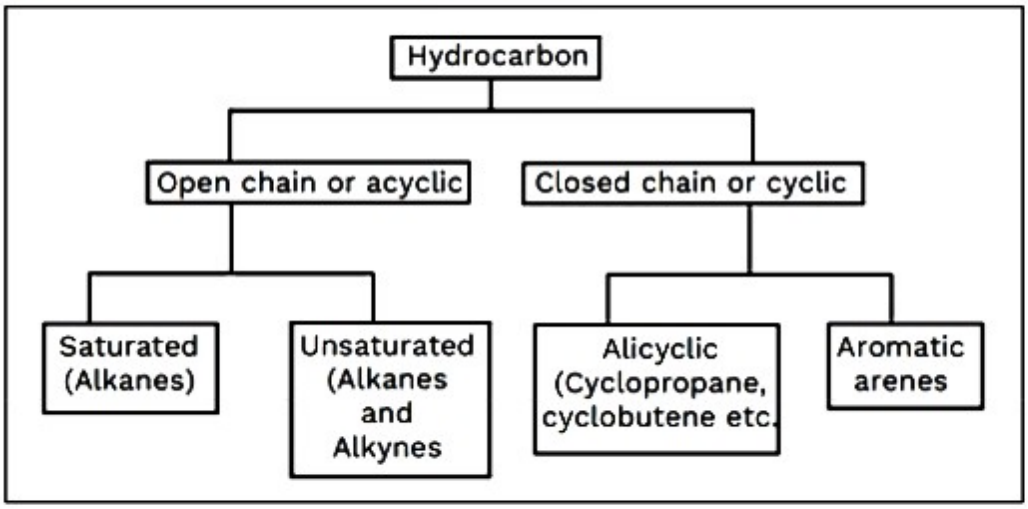
(i) 

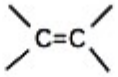
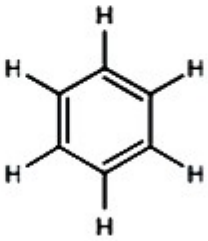
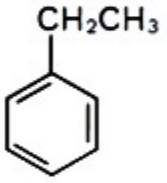
(j) 

(k) 

Hydrocarbon

Hydrocarbons are Broadly Classified as:

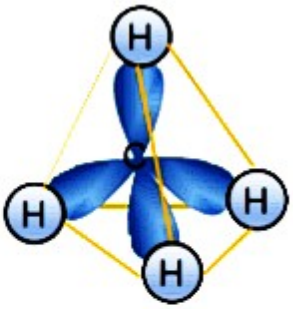


Hydrocarbon	Functional group	Example
Alkanes	none (no double or triple bonds)	$CH_3-CH_2-CH_3$, propane
Alkenes	 double bond	$CH_2=CH-CH_3$, propene
Alkynes	$-C\equiv C-$ triple bond	$H-C\equiv C-CH_3$, propyne
Aromatics	 Benzene Ring	 ethylbenzene

Alkanes

(i) The alkanes (paraffins) are the saturated hydrocarbons. Many occur naturally, and the main source of the alkanes is mineral oil or petroleum, which found in many parts of the world.

These are also called as 'Paraffins'(Parum+Affinis i.e. less reactive).



(ii) General formula is C_nH_{2n+2}.

(iii) Hybridisation state of carbon is sp^3.

(iv) Geometry of carbon is tetrahedral.

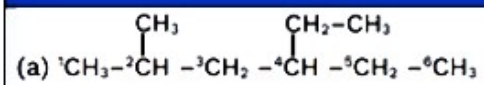
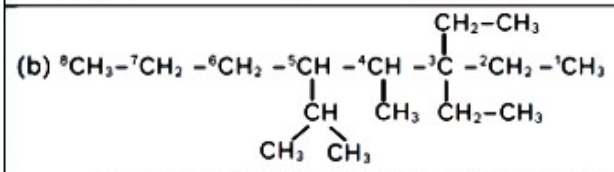
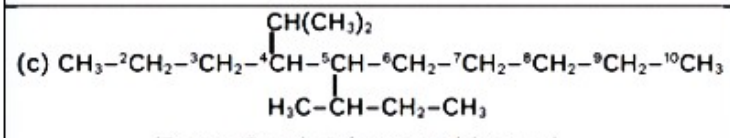
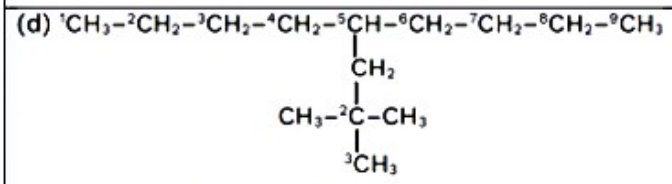
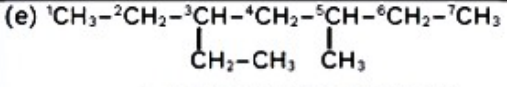
(v) Bond angle is 109°28'.

(vi) Number of bond angle in methane are six, while in ethane are twelve.

(vii) C – C bond length is 1.54 Å while C–H bond length is 1.11Å.

(A) Nomenclature

Nomenclature of a few organic Compounds:

Structure and IUPAC name	Remarks
 (4-Ethyl-2-methylhexane)	Lowest sum and alphabetical arrangement
 (3, 3-Diethyl-5-isopropyl-4-methyloctane)	Lowest sum and alphabetical arrangement
 (5-sec-Butyl-4-isopropyldecane)	Sec is not considered while arranging alphabetically; isopropyl is taken as one word
 5-(2,2-Dimethylpropyl)nonane	Further numbering to the substituents of the side chain
 3-Ethyl-5-methylheptane	Alphabetical priority order

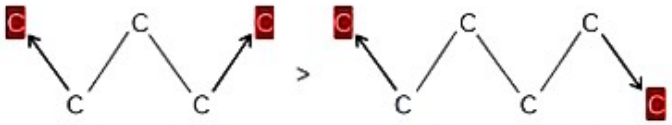
(B) Physical Properties

(i) (a) C_1 to C_4 ⇒ No colour, No odour
(b) C_5 to C_{17} ⇒ colourless & odourless liquid
(c) Rest are colourless solids.

(ii) Non polar molecules, not soluble in polar solvent like water. With-increase in size of hydrophobic part solubility decreases.

(iii) MELTING AND BOILING POINTS

Increases with molecular mass and decreases with No. of branches. As far as melting point is concerned the alkane having even carbons has more M.P. than odd carbons, since the intermolecular forces in a crystal depend not only upon the size of the molecules but also upon a crystal lattice.



(both methyl group are directed on same side) > (both methyl group are adversely directed)

n-pentane n-hexane

M.P. [even > odd no. of atoms]

(iv) Boiling point decreases with the increment of branches.

(v) Physical state:
Alkanes
C_1– C_4 Gaseous state
C_5 – C_{17} Liquid state
(except neo-pentane)
C_{18} & above Solid like wax

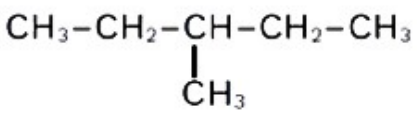
(vi) Alkanes are lighter than water, so it floats over water.

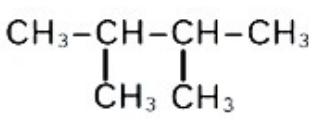
Concept Building Exercis

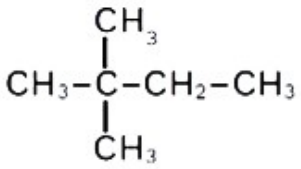
Q36 **For alkanes having molecular formula C_6H_{14}, write structures of different chain isomers. Also write their IUPAC names.**

Sol. (i) $CH_3-CH_2-CH_2-CH_2-CH_2-CH_3$: n-Hexane

(ii) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-CH_2-CH_3$ 2-Methylpentane

(iii)  3-Methylpentane

(iv)  2,3-Dimethylbutane

(v)  2,2-Dimethylbutane

Q37 **Write IUPAC names of the following compounds :**
(i) $(CH_3)_3$ C $CH_2C(CH_3)_3$
(ii) $(CH_3)_2$ $C(C_2H_5)_2$

Sol. (i) 2, 2, 4, 4-Tetramethylpentane
(ii) 3, 3-Dimethylpentane

Q38 **In the above structures (i-iv), Count the number of sigma (σ) and pi (π) bonds.**

Sol. σ bonds : 33 π bonds : 2
σ bonds : 17 π bonds : 4
σ bonds : 23 π bonds : 1
σ bonds : 41 π bonds : 1

Alkene

1. General Introduction

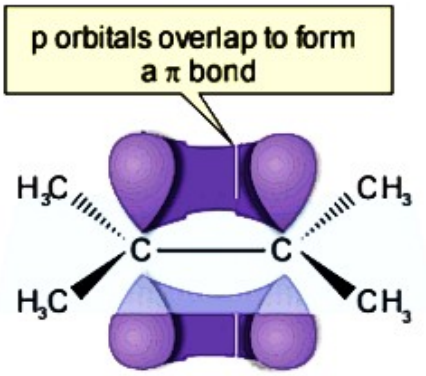
Hydrocarbons that contain a carbon–carbon (C=C) double bond are called alkenes.
Each double-bonded carbon(C=C) of of an alkene has three sp^2 orbital of another atom to form a σ bond.
Thus, one of the carbon–carbon bonds (C-C) in a double bond is a σ bond, formed by the overlap of an sp^2 orbital of one carbon with an sp^2 orbital of the other carbon.

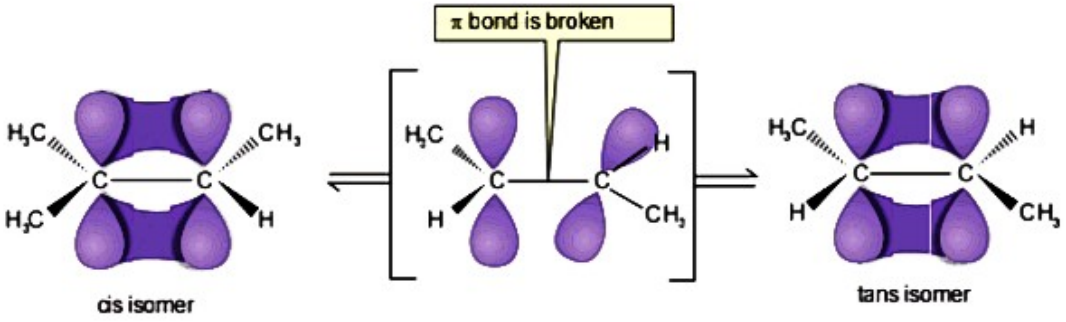
The second carbon-carbon(C-C) bond in the double bond (the π bonds) is formed from side-to-side overlap of the remaining p-orbitals on the sp^2 carbons. Because three points determine a plane, each sp^2 hybridized carbon and two of the three atoms bonded to it are in a plane.

In order to achieve maximum orbital-orbital overlapping, the two p-orbitals must be parallel to each other. So that, all six atoms of the double bond system are in the same plane. A molecular orbital description of a carbon-carbon double bond.

It is important to remember that the π bond represents the electrons cloud that is above and below the plane defined by the two sp^2 hybridised carbons and the four atoms bonded to them.

Because the two p-orbitals that form the π bond must be parallel to one another to attain maximum overlap, and rotation about a double bond does not occur. If rotation about a double bond did occur, the two p-orbitals would ceases to overlap and the π bond would be destroyed.





2. Characteristics of Alkene

(a) General formula : C_nH_{2n}.

(b) Functional group : in alkenes is > C = C <

(c) The double bond(C=C) is made up of one sigma and one pi bond.

(d) The doubly bonded carbon atoms are sp^2 hybridized.

(e) Geometry of unsaturated 'C' carbon is trigonal planar.

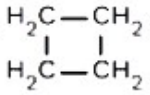
(f) C=C bond length is 1.34 Å.
(g) C–H bond length is 1.10 Å.
(h) C=C bond energy is 143.1 k cal mol^{-1}.
(i) C–H bond energy is 98.69 k cal mol^{-1}.
(j) Alkene shows chain, position, functional (ring chain), optical & geometrical isomerism.

Cumulated polyene can exhibit geometrical as well as optical isomerism. for example. Isomers of C_4H_8:

(i) Chain isomers → $CH_3-CH_2-CH=CH_2$ & $H_3C-C(CH_3)=CH_2$

(ii) Position isomers → $CH_2=CH-CH_2-CH_3$ & $CH_3-CH=CH-CH_3$

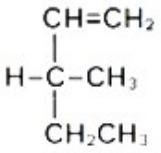
(iii) Functional isomers (ring chain isomerism)

$CH_3-CH_2-CH=CH_2$ & 

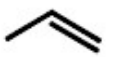
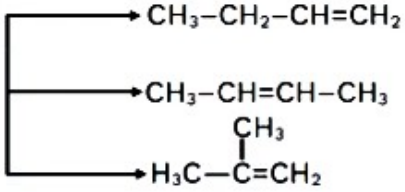
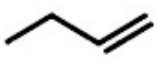
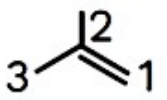
1-butene cyclobutane

(iv) Geometrical isomers → 

cis-2-butene cis-2-butene

(v) Optical isomerism → 

3. Structure, IUPAC name, and Common Name of Some Alkenes

Stick	Structure	IUPAC name	Common or systematic or trivial name
$CH_2=CH_2$		Ethene	Ethylene
$CH_3-CH=CH_2$		Propene	Propylene
		But-1-ene	1-Butylene
		But-2-ene or 2-butene	2-Butylene
		2-Methyl prop-1-ene	Isobutylene or isobutene

4. Physical Properties

(i) From C_2–C_4 they are colourless, odourless gases, from C_5–C_{17} they are colourless liquids, C_{18} onwards alkenes are solids.

(ii) Alkenes are practically insoluble in water because they cannot form hydrogen bonds with H_2O molecule. They dissolve freely in organic solvents like benzene, chloroform, CCl_4 petroleum ether, etc. (Like dissolves like)

(iii) If branching increases, boiling point and melting point both decline.

(iv)



(i) (ii)

M.P. (ii) > (i) (Symmetry)

B.P. (i) > (ii) (Polarity)

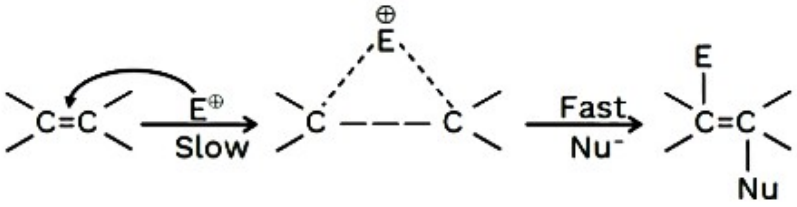
(v) The boiling points of cis isomers are higher than trans isomers because cis–alkenes has greater polarity (Dipole moment) than trans one.

(vi) Alkenes floats over water.

(vii) The increase in branching in carbon chain decreases the boiling point among isomeric alkenes.

5. Chemical Properties

The main reactions of alkene are electrophilic addition reaction and the mechanism is based on ionic. It was proved by Francis.



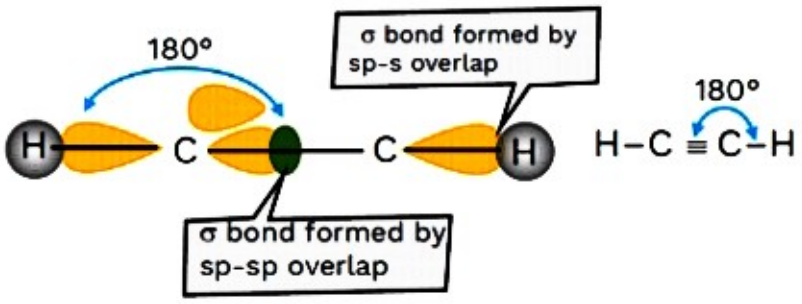
Alkyne

1. General Introduction

(i) Similar to alkene, having one 'σ' and 2'π' bonds.

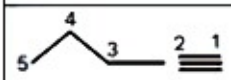
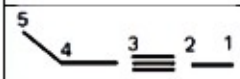
(ii) Other term used for alkyne is acetylenes considering derivatives of Ethyne (acetylene), the simplest hydrocarbon

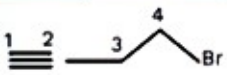
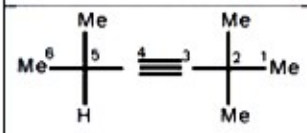
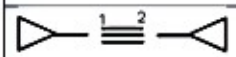
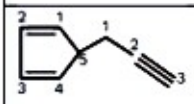
(iii) Industrial significance of acetylene (Ethyne) is welding purpose. Colorless, foul-smelling gas famous in domestic and commercial use. When acetylene flame (sooty flame) is supplied with pure oxygen, color turns to light blue and flame temperature increases dramatically.



(iv) Bond angle in alkyne is 180º.
(v) Their general formula is C_nH_{2n-2}
(vi) C–C triple bond length is 1.20 Å.
(vii) C–H bond length is 1.08 Å.

2. IUPAC and Common Name of Some Acetylenes and Its Derivate

Structure	IUPAC name	Common name
	Ethyne	Acetylene
	Pent-1-yne	Propylacetylene
	Pent-2-yne	Ethyl methylacetylene

Structure	IUPAC name	Common name
	4-Bromobut-1-yne	2-Bromoethyl acetylene
	2,2,5-Trimethyl hex-3-yne	tert-butyl isopropyl acetylene
	1,2-Dicyclopropyl eth-1-yne (Numbers are not necessary, but are used for clarity)	Dicyclopropyl acetylene
	5-(2-Propynyl)-cyclopenta-1,3-diene	-

3. Physical Properties

(a) Colourless, odourless and tasteless hydrocarbon.

(b) Polarization power in case of lower alkyne (example C_2H_2), lead and partial solubility in H_2O.

(c) With increase in size of hydrophobic part, solubility decreases dramatically.

(d) Solubility in organic solvent.

(e) Melting point ↑ M.W. ↑ Branching ↓ (consider MP or BP increasing)

Boiling point ↑ M.W. ↑ Branching ↓ (consider MP or BP increasing)

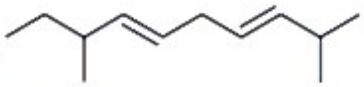
(f) C_1–C_4 Gaseous.
C_5–C_{11} Liquid.
C_{12} & up Solid.

4. Chemical Properties

The chemical behaviour of alkynes is similar to that of alkenes. Alkynes form addition products with two or four univalent atoms or groups. They are generally less reactive than alkenes towards electrophilic addition reactions even though π electron density is higher in alkynes. This is because on moving from alkene to alkyne, the C—H bond has more of s character (33% in alkenes as compared to 50% in alkynes). Therefore, the π-electrons are more firmly held by carbon nuclei in alkynes and are thus less reactive to an electrophile.

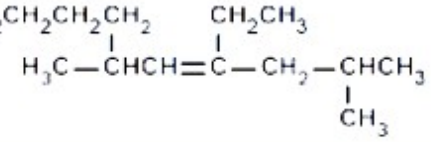
Concept Building Exercis

Q39 **Write IUPAC names of the following compounds :**

(i) 

(ii) 

(iii) $CH_2 = C(CH_2CH_2CH_3)_2$

(iv) 

Sol. (i) 2, 8-Dimethyl-3, 6-decadiene;
(ii) 1, 3, 5, 7 Octatetraene;
(iii) 2-n-Propylpent-1-ene;
(iv) 4-Ethyl-2,6-dimethyl-dec-4-ene

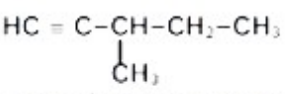
Q40 **Write IUPAC names of all the isomers corresponding to the 5th member of alkyne series also draw structures different isomers. What kind of isomerism is exhibited by different pairs of isomers?**

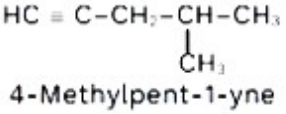
Write possible isomers of 5th member of alkyne has the molecular formula C_6H_{10}.

(a) $HC \equiv C-CH_2-CH_2-CH_2-CH_3$
Hex-1-yne

(b) $CH_3-C \equiv C-CH_2-CH_2-CH_3$
Hex-2-yne

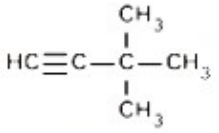
(c) $CH_3-CH_2-C \equiv C-CH_2-CH_3$
Hex-3-yne

(d) 
3-Methylpent-1-yne

(e) 
4-Methylpent-1-yne

(f) $CH_3-C\equiv C-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

4-Methylpent-2-yne

(g) 

3,3-Dimethylbut-1-yne

Application of Stereochemistry

1. Photo halogenation of Alkane

Substrate	→	Alkane
Reagent	→	X_2 / hν or X_2 / Δ
Intermediate	→	Carbon free radical
Mechanism	→	Free radical substitution reaction
Product	→	Haloalkane (Mono halogenated, Di halogenated, etc.)
Reaction	→	$R-H \xrightarrow{X_2/h\nu} R-X+H$

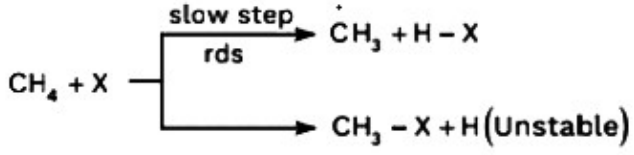
Mechanism

(i) Chain initiation Step



(Always endothermic)

(ii) Chain Propogation Step



Step-1

Step-2 $\dot{C}H_3 + X-X \rightarrow CH_3-X+X^{\bullet}$

(may be endothermic or exothermic)

(iii) Chain Terminating Step

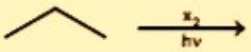
(Always exothermic ΔH = negative)

$CH_3^{\bullet} + CH_3^{\bullet} \longrightarrow CH_3-CH_3$ (Dimer)

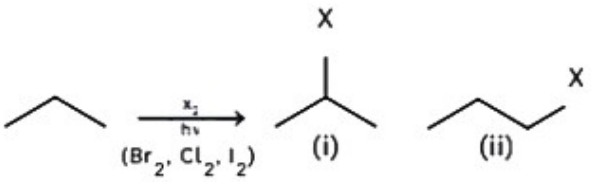
$CH_3^{\bullet} + X^{\bullet} \longrightarrow CH_3-X$ (Desired product)

$X^{\bullet} + X^{\bullet} \longrightarrow X_2$

ILLUSTRATION

Q41 

Sol.



Characteristics of Reaction

1. **Rate of photo halogenation :**

   Rate of reaction ∝ Stability of carbon free radical

   Stability of carbon free radical : 3° > 2° > 1°

   Rate of reaction ∝ Bond strength of H–X

   Bond strength of H–X : H–F > H–Cl > H–Br > H–I

   Rate of Photo halogenation: $F_2 >>> Cl_2 > Br_2 >>>> I_2$
2. Direct fluorination is not possible as it is explosive, so fluorination takes place in dark.
3. Iodination is reversible reaction as in this reaction strong reducing agent HI is formed which reduces alkyl iodide into alkane again. So in iodination we use strong oxidizing agent like HNO_3, HIO_3 etc. So that reaction move in forward direction.
   (Follows Le-chatelier principle)

   $R-H \underset{}{\overset{I_2/h\nu}{\rightleftharpoons}} R-I + HI$

   $5HI + HIO_3 \longrightarrow \underset{\text{(Reused)}}{\boxed{3I_2}} + 3H_2O$

   (Follows Le-chatelier principle)

   Reaction move in forward direction.

ILLUSTRATION

Q42 $CH_4 \xrightarrow{Cl_2,\ h\nu}$

Sol. $CH_4 \xrightarrow{Cl_2,\ h\nu} CH_3-Cl + CH_2-Cl_2 + CH-Cl_3 + CCl_4$

Q43 CH_4 (excess) + $Cl_2 \xrightarrow{h\nu} CH_3Cl$

Think about role of excess CH_4 ?

Sol. Excess CH_4 will consume all Cl_2 so no further chlorination.

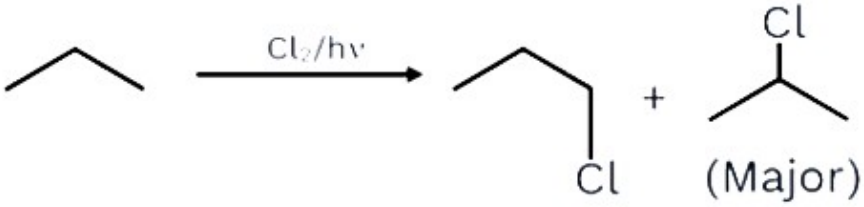
Q44 $CH_4 + Cl_2$ (excess) $\xrightarrow{h\nu}$ Major product

Sol. $CH_4 + Cl_2$ (excess) $\xrightarrow{h\nu} CCl_4$

Q45 $CH_3 - CH_3 \xrightarrow[\text{Excess}]{Cl_2, h\nu}$ Number of possible product

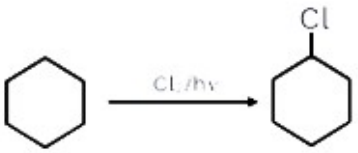
Sol. $CH_3 - CH_3 \xrightarrow[\text{Excess}]{Cl_2, h\nu} CH_3{-}CH_2Cl + CHCl_2{-}CCl_3 + CH_2Cl{-}CH_2Cl + CH_3{-}CHCl_2 +$
$CH_3{-}CCl_3 + CHCl_2{-}CHCl_2 + CCl_3{-}CCl_3 + CH_2Cl{-}CCl_3 + CH_2Cl{-}CHCl_2$

Q46  $\xrightarrow{Cl_2/h\nu}$ Number of monohalogenated product

Sol. 

Q47  $\xrightarrow{Cl_2/h\nu}$ Number of monohalogenated product

Sol.



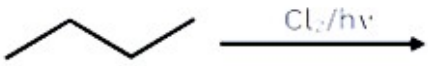
Q48



Sol.

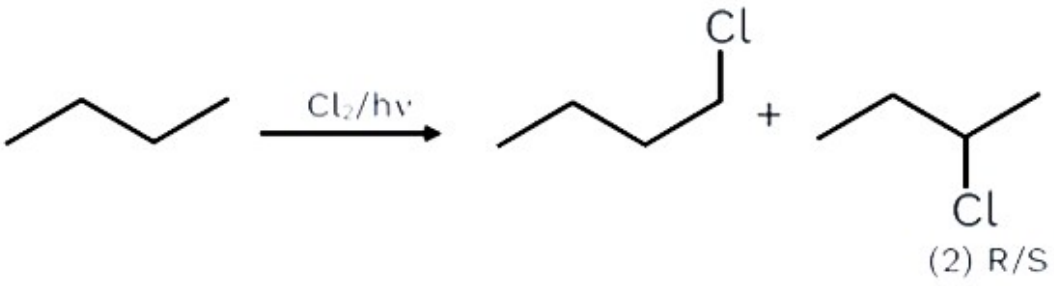


Q49



(a) Total number of monochlorinated product =
(b) Enantiomer pair =
(c) Enantiomer =
(d) Number of product separated by fractional distillation =
(e) Number of chiral product =
(f) Number of structural product =

Sol.



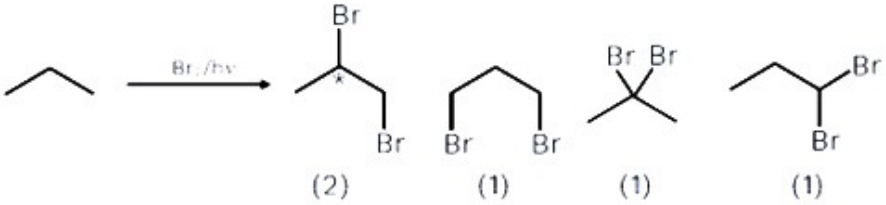
(a) Total number of monochlorinated product = 3
(b) Enantiomer pair = 1
(c) Enantiomer = 2
(d) Number of product separated by fractional distillation = 2
(e) Number of chiral product = 2
(f) Number of structural product = 2

Q50



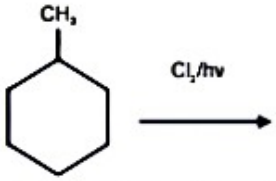
Number of dihalogenated product

Sol.



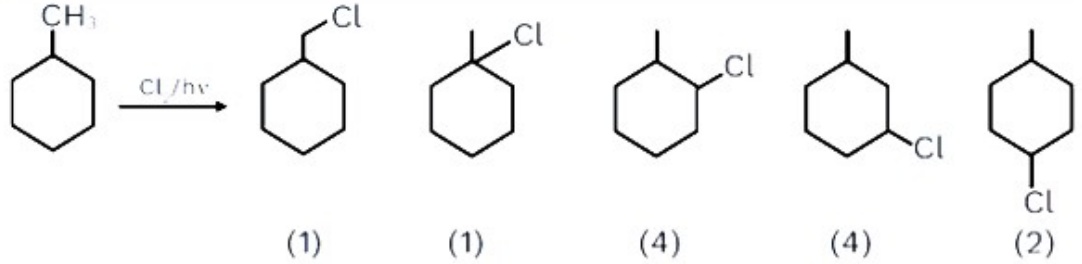
(i) Total number of product = 5
(ii) Number of product separated by fractional distillation = 4

Q51



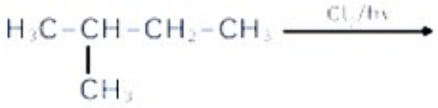
(i) Total number of monohalogenated product =
(ii) Total number of structural monohalogenated product =
(iii) Total number of product separated by fractional distillation =
(iv) Enantiomer pair =
(v) Chiral =
(vi) Achiral =

Sol.



(i) Total number of monohalogenated product = 12
(ii) Total number of structural monohalogenated product = 5
(iii) Total number of product separated by fractional distillation = 8
(iv) Enantiomer pair = 4
(v) Chiral = 8
(vi) Achiral = 4

Q52



Mono halogenation

(i) Total number of monohalogenated product =

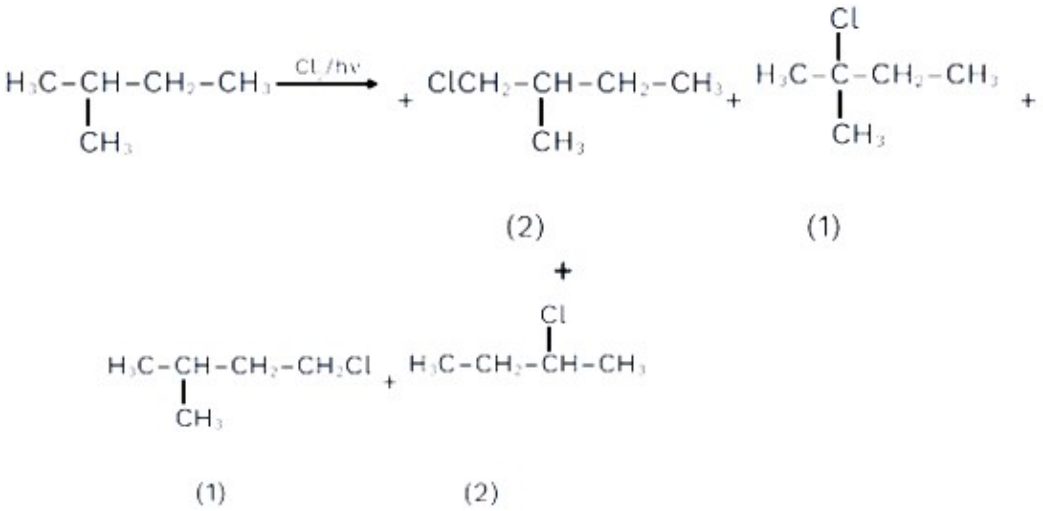
(ii) Total number of structural monohalogenated product =

(iii) Total number of product separated by fractional distillation =

(iv) Enantiomer pair =

(v) Chiral =

(vi) Optical Inactive =



Sol.

(i) Total number of monohalogenated product = 6

(ii) Total number of structural monohalogenated product = 4

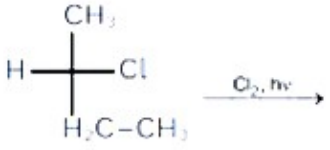
(iii) Total number of product separated by fractional distillation = 4

(iv) Enantiomer pair = 2

(v) Chiral = 4

(vi) Optical Inactive = 2

Q53

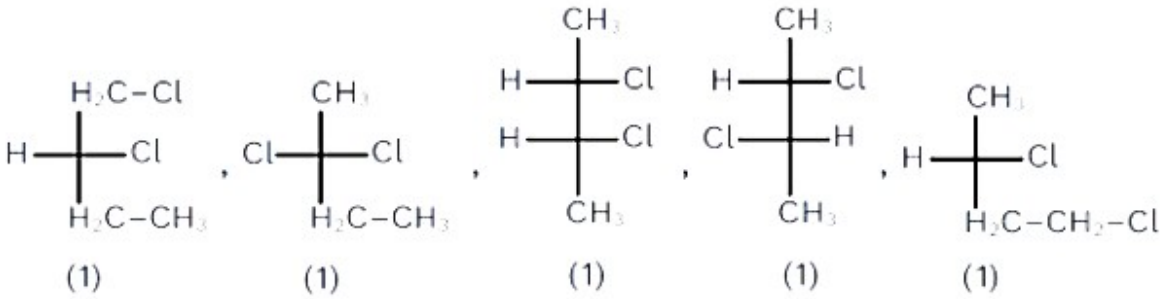


(i) Total number of monohalogenated product

(ii) Total number of structural monohalogenated product

(iii) Total number of product separated by fractional distillation

Sol.



(1) (1) (1) (1) (1)

(i) Total number of monohalogenated product = 5
(ii) Total number of structural monohalogenated product = 5
(iii) Total number of product separated by fractional distillation = 5

How to decide Major and Minor product in Photo halogenation

X_2	Relative Rates		
	1°H	2°H	3°H
F_2	1	1	1
Cl_2	1	3.5	5
Br_2	1	82	1600

$$\text{Selectivity} \propto \frac{1}{\text{Reactivity}}$$

Reaction of product = Relative rate of H × Number of that H

ILLUSTRATOIN

Q54 $CH_3 - CH_2 - CH_3 \xrightarrow{F_2/h\nu}$

Sol.

$$CH_3 - CH_2 - CH_3 \xrightarrow{F_2/h\nu} \underset{\underset{X\%}{\underset{25\%}{}}}{CH_3 - \underset{|}{\overset{F}{CH}} - CH_3} + \underset{\underset{y\%}{\underset{75\%}{}}}{CH_3 - CH_2 - \underset{|}{\overset{F}{CH_2}}}$$

1° > 2°

Q55 **Number of monochloro products :**
(a) Including stereoisomers
(b) Excluding stereoisomers

$H_3C-CH_2-CH_3 \xrightarrow{Cl_2/h\nu}$

Sol. (a) 2

(b) 2

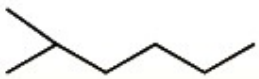
$H_3C-CH_2-CH_3 \xrightarrow{Cl_2,\ h\nu} H_3C-\underset{Cl}{CH}-CH_3 + H_3C-CH_2-\underset{Cl}{CH_2}$

$2 \times 3.5 = 7$ $\qquad$ $6 \times 1 = 6$

$\frac{7}{13} \times 100$ $\qquad$ $\frac{6}{13} \times 100$

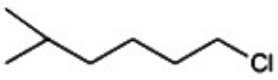
53.8% $\qquad$ 46.1%

$2^\circ > 1^\circ$

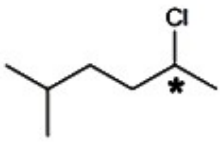
Q56 

Sol. 10, 6

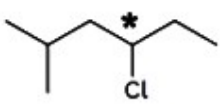
Number of monochloroproduct = 10 $\qquad$ Number of monochloroproduct = 6 (excluding stereoisomer)



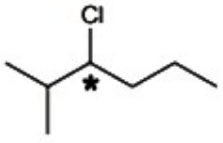
$\Rightarrow 1$



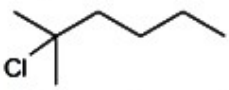
$\Rightarrow 2$



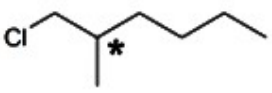
$\Rightarrow 2$



$\Rightarrow 2$

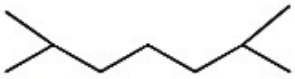


$\Rightarrow 1$



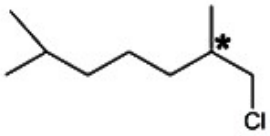
$\Rightarrow 2$

Q57

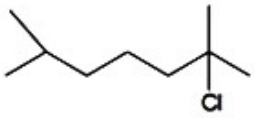


Sol. 6, 4

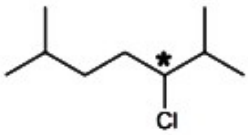
Number of monochloroproduct = 6 = 4 (excluding stereoisomer)



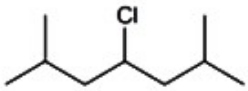
⇒ 2



⇒ 1



⇒ 2



⇒ 1

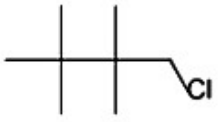
Q58

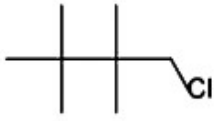


Sol. 1, 1

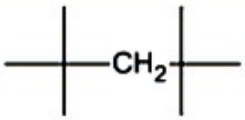
Number of monochloroproduct = 1

Number of monochloroproduct = 1 (excluding stereoisomer)





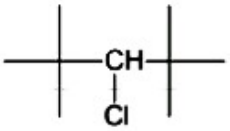
Q59

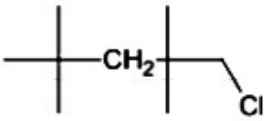


Sol. 2, 2

Number of monochloroproduct = 2

Number of monochloroproduct = 2 (excluding stereoisomer)



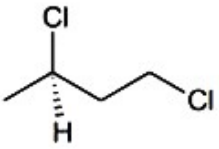


Q60 **R-2-chlorobutane**

Sol. 5, 3

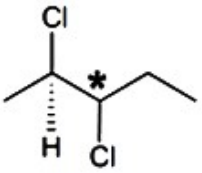
Number of Dichloroproduct = 5

Optically active product = 4 (including stereoisomer)



⇒ 1

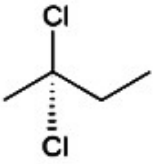
⇒ Optically active



⇒ 2

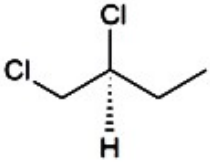
i.e. One optically active

One meso compound



⇒ 1

⇒ Optically active



⇒ 1

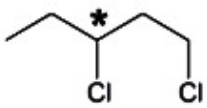
⇒ Optically active

Q61 3-chloropentane

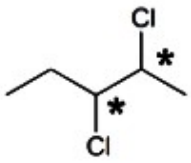
Sol. 7, 6

Number of Dichloroproduct = 7 Optically active product = 6

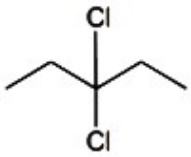
(including stereoisomer)



⇒ 2 Optically active ✓



⇒ 4 Optically active ✓



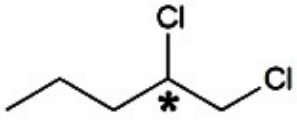
⇒ 1 **Optically inactive**

Q62 R & S-2-chloropentane

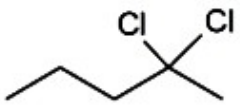
Sol. 12, 10

Number of Dichloroproduct = 12 Optically active product = 10

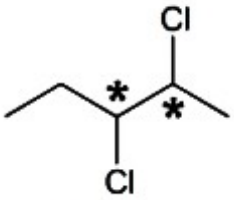
(including stereoisomer)



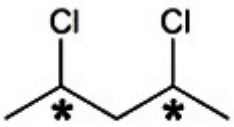
⇒ 2



⇒ 1

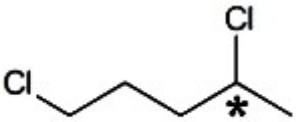


⇒ 4



⇒ 3

i.e. Two optically active
One meso compound



⇒ 2

Reaction of Alkane with SO_2Cl_2/hv: (Reed Reaction)

Points to remember

$$R - H \xrightarrow{SO_2Cl_2} R - SO_2 - Cl$$

Chloro Sulphonation

$$C_3H_8 + SO_2 + Cl_2 \xrightarrow{uv} \underset{\text{propane sulphonyl chloride}}{C_3H_7SO_2Cl} + HCl$$

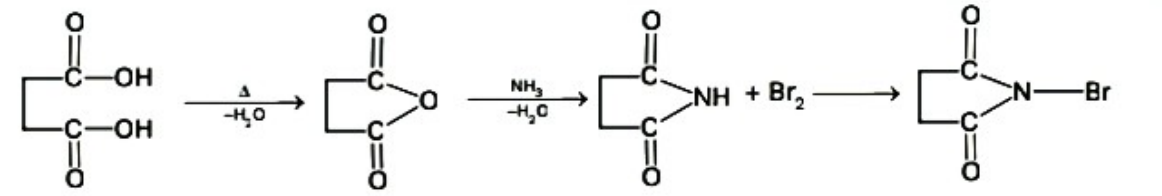
Reaction with NBS (N-Bromo succinimide)/hv:

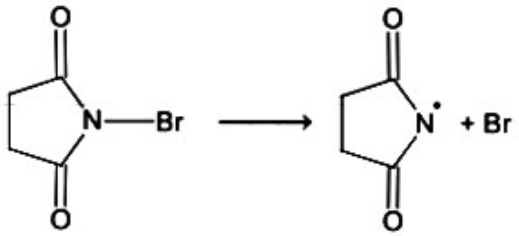
Substrate	:	Alkene, Benzene (Aromatic compound)
Reagent	:	NBS / hv
Intermediate	:	Carbon free radical
Mechanism	:	Free radical substitution reaction
Product	:	Allylic and Benzylic substituted bromide

Reaction:

(i) $CH_2 = CH - CH_3 \xrightarrow{NBS/hv} H_2C = CH - \underset{Br}{CH_2}$

(ii) 





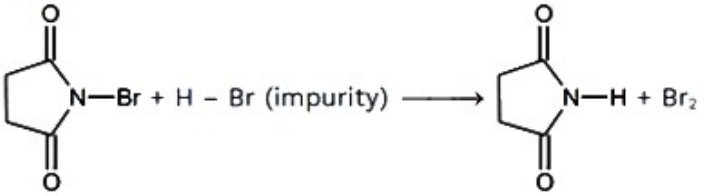
Chain Propagation Step

$$CH_2 = CH - CH_3 + \dot{B}r \xrightarrow[\text{slow}]{\text{rds}} CH_2 = CH - {}^{\cdot}CH_2 + HBr$$

$$CH_2 = CH - CH_2 + Br - Br \longrightarrow CH_2 = CH - CH_2 - Br + Br$$

$$Br^{\cdot} + Br^{\cdot} \longrightarrow Br_2 \text{ (Chain terminating step)}$$

Note: Along with NBS some amount of HBr is also used to maintain the proper concentration of Br_2.



Method to Calculate Structural Product in this Reaction

Step-1 : First identify allylic or benzylic carbon in the given substrate.
Step-2 : By the removal of H-atom at this allylic or benzylic position convert into free radical.
Step-3 : Draw all possible resonating structure for allylic or benzylic position except in benzene ring.
Step-4 : Add bromine radical to all possible position of carbon radical formed.

ILLUSTRATIONS

Q63 $CH_3 - CH = CH - CH_3 \xrightarrow[h\nu]{NBS}$

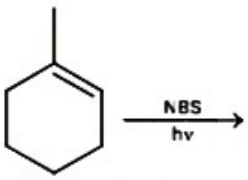
(A) Total number of products (x) = ?
(B) Total number of structural products (y) = ?
(C) Total number of products separated by fractional distillation (z)=?
(D) Chiral product = ?
(E) Optically Inactive = ?

Sol. $CH_3 - CH = CH - CH_3 \xrightarrow[h\nu]{NBS} CH_3 - CH = CH - CH_2 - Br + CH_3 - \underset{\underset{Br}{|}}{CH} - CH = CH_2$

(2) Geometrical isomer (2) Optical isomer

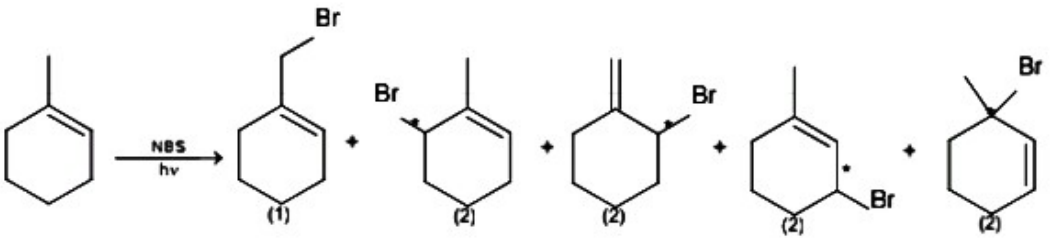
(A) Total number of products (x) = 4
(B) Total number of structural products (y) = 2
(C) Total number of products separated by fractional distillation (z)=3
(D) Chiral product = 2
(E) Optically Inactive = 2

Q64



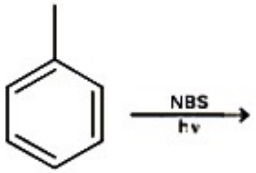
(a) Total number of products (x) = ?
(b) Total number of structural products (y) = ?
(c) Total number of products separated by fractional distillation (z)=?

Sol.



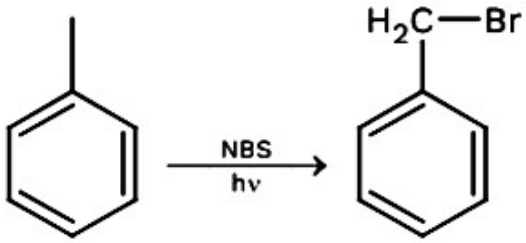
(a) Total number of products (x) = 9
(b) Total number of structural products (y) = 5
(c) Total number of products separated by fractional distillation (z)=5

Q65



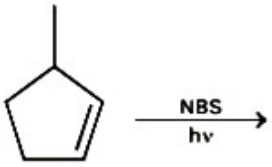
(a) Total number of products (x) = ?
(b) Total number of structural products (y) = ?
(c) Total number of products separated by fractional distillation (z)=?

Sol.



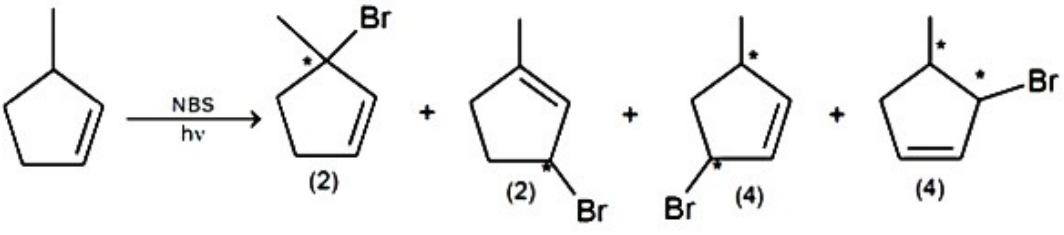
(a) Total number of products (x) = 1
(b) Total number of structural products (y) = 1
(c) Total number of products separated by fractional distillation (z)=1

Q66



(a) Total number of products (x) =
(b) Total number of structural products (y) =
(c) Total number of products separated by fractional distillation (z)=

Sol.



(a) Total number of products (x) = 12
(b) Total number of structural products (y) = 4
(c) Total number of products separated by fractional distillation
(z) = 6